

Design and Development of Interactive LED Display for Fan Applications

Jui-Hung Cheng ^{1*}, Hsin-Hung Lin ²

¹ Department of Mold and Die Engineering, National Kaohsiung University of Applied Science
No. 415, Jiangong Rd., Sanmin Dist., Kaohsiung 80778, Taiwan
E-mail: rick.cheng88@gmail.com

² Department of Creative Product Design, Asia University
No. 500, Lioufeng Rd., Wufeng, Taichung 41354, Taiwan
E-mail: hhlin@asia.edu.tw; a123lin0@gmail.com

Abstract

This paper describes how to apply the external driving motor structure to trigger the inner motor structure of the impeller, and self-generate DC power through the conversion of electromagnetic energy, which has the advantages of simple structure, reduced fan thickness and no need of another extra power to provide for the LED display circuit. Moreover, with the contactless capacitive touchpad, to demonstrate the new interactive interface design, showing the dynamic text and graphics information, etc. The purpose is to integrate new motor structure, supplemented by the relevant LED and sensor applications to enhance the added value of the traditional cooling fan products, and diversified applications with creative digital content, can be used as a reference for industries upgrading and product design.

Key words: Contactless Power Transfer, External Driving Motor, Interactive Display, LED Fan, Product Design

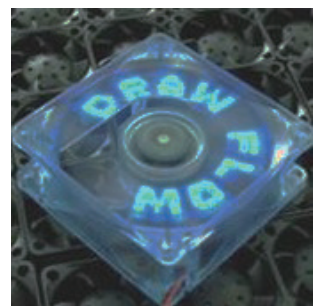
Introduction

In recent years, the development of miniaturized and multi-functional high-power electronic products, if not effectively and rapidly dissipating heat sources, will result in abnormal operation of the product system. Usually a cooling fan is used to generate air-flow and air-pressure through the motor-driven impeller to achieve the purpose of cooling. Due to the driving motor was installed inside of the rotor impeller, this kind of small BLDC motor structure is hard to wire coiling, and not easy cool the temperature of motor by itself. When the impeller rotates, the pressure between the top of the blade and the outer frame suddenly increases, resulting in the disadvantages of greater noise. [1]. These noise sources may come from the electromagnetic force, mechanical operation, air flow and pressure, power supply, etc. As the motor is running, the angular position of the rotor between the stator and the rotor, as well as the change of the length of the air gap, can also cause vibration [2]. Therefore, by the analysis of magnetic flux distribution, the flux density of air gap between rotor and stator, the back electromotive force, and obtain the optimal design arc length and angle of north and south poles to reduce vibration of back EMF [3]. This type of motor structure design usually not effectively achieve the heat dissipation for large area applications, need to be

assisted by other thermal components, such as heat pipes, heat sinks, etc. This will complicate the design of the cooling system and increase the cost. In electronic products toward miniaturization and high efficiency trends, especially the permanent magnet brushless DC motor (PM BLDC) products such as small size, light weight, high efficiency, and reliability and so on, the relevant product applications and developments have great potential and opportunities. For the LED technology in the mature market and cost reduction, many products are combined with LED for new types of product design. While the traditional fan with LED module to design the texts and graphics display above the impeller, usually must add extra DC power or increase one motor structure to generate power through the reverse operation (Fig. 1). However, this design has some problems such as noise, running life, and complex structure, etc. In the text and graphic dynamic display, only in accordance with the pre-burning program to render, unable to meet the user's desired touch mode and instantly change function. To meet the diversified market demand, this study applies the principle of visual retention, different from the external multi-pole motor design, by the single pole motor to drive the cooling fan, and reverse the internal motor of the impeller to be a power generator to supply DC power for LED display to show text and graphic and other information. This approach can enhance the product values and enhance the competitiveness in niche markets.



LED module on the impeller



Text and graphic display effects

Fig. 1 Traditional LED display fan.

External Driving Motor

Traditional external permanent magnet motor (permanent parochial, PM) structural characteristics, mainly to improve the energy density, the motor structure is compact through the magnetic field calculated under the pole width, thickness, helix angle design, can effectively cool the motor. Moreover, the stator core in the cutting of multiple air channels can also increase air flow in order to cool the magnet [4]. The requirements of motor products must have low noise; therefore, the cooling fans mainly use brushless DC motor. In terms of space saving, the compact axial fan products can be driven by the back electromotive force, cogging torque, axial preload and other analysis for miniaturization design, increase static pressure and air volume, and vibration suppression, will be the next generation of mobile equipment cooling solution [5]. To further reduce the structural design, apply the contactless driving technology that through the external rotor motor structure, especially the tip-magnetic driving motor is very effective. Under the equivalent magnetic circuit analysis, this design is different from the traditional fan structure, its motor rotor, stator, and blade positions, the arrangement makes the fan motor has a higher stall torque, better efficiency, and significantly lower fan noise [6]. In addition, under the consideration of energy saving, the advantage of using the tip- magnetic driving motor design, moving the impeller hub from center out to external to increase the length of the blade and expand the heat dissipation area. Moreover, with higher motor torque, low noise advantages, and enhances the overall fan cooling efficiency [7]. For example, the Tip-Magnetic Driving Fan (TMD Fan) mainly in the fan-blade internal rotor structure moved outside the blade top edge, the blade above the hard magnet ring, divided into 12 poles around, and the internal motor stators moved out to four fan frame corners [8].

Contactless Power Transfer

The magnet mechanism composed of magnets and coils is designed to convert mechanical energy into electric energy to realize the application of contactless power transmission. There are mainly two kinds of research in related researches, one is the swing type and the other is rotary motion. In the swing type, the electromagnetic force of the piezoelectric material is mainly used to set the electromagnet in two elastic slices, oscillating back and forth to swing the fan blades for cooling. Compared with the rotary fan, it has the advantages that there is no motor and bearing, the structure is simple, but for most applications it only can be driven under low power (1.74 W) for most applications and has limited heat dissipation [9]. In the application of rotary fans, the rotary contactless power transmission system is through the high coupling capacity of magnetic materials, and brushless bidirectional feedback motor structure, rotating induction coupling structure to enhance transmission power and stability [10]. Through the above two types of motor concept, this study applied the external driving motor structure to generate a contactless and stable power source, and to

provide a high-efficiency and low-cost power transmission system. An external magnetic motor is used to drive the internal impeller rotation (Fig. 2), which has an inner motor to interact with the magnetic strips fixed on the enclosure to generate the alternating current. Then through the rectifier filter circuit, the AC voltage is converted into DC voltage, provide power to the display circuit of LED PCBA on the impeller (Fig. 3). And the microcontroller unit (MCU), per the written firmware program, the implementation of text and graphic display, as well as the action setting mode.

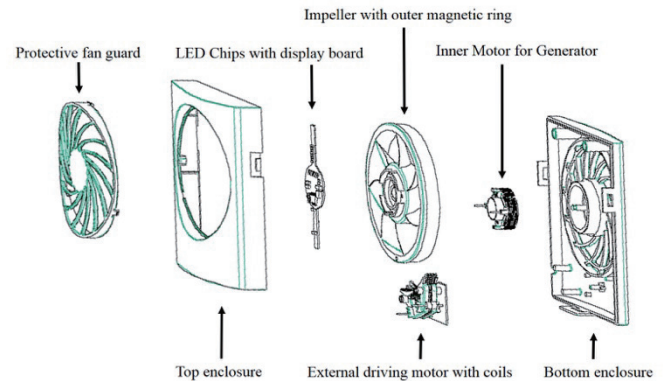


Fig. 2 Contactless touchpad with external motor structure.

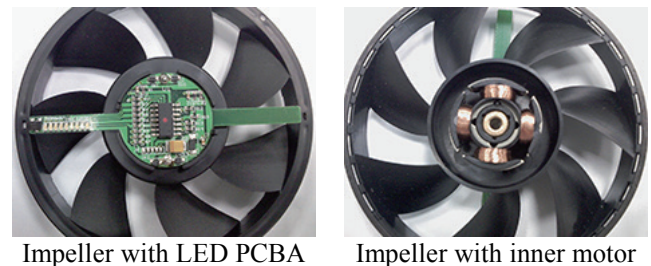


Fig. 3 Impeller prototype with LED PCBA & inner motor.

Electrical Design for Interactive Display

If a DC motor is implemented via contacts or contactless measurement of speed for monitoring temperature and other related values, it is required to use additional equipment. If spatial and structural limitations exist, sometimes contactless optical sensors will be used when the cable measurements are not available. However, if the larger vibration of the fan rotor or has eccentric operation, will affect the measurement inaccuracy. In some special environments, such as measuring the temperature change caused by moving vehicles in a tunnel, the sensors were placed in a tube-type ventilator to automatically detection and display numerical changes and regulated [11]. Through the sensors, the energy conversion and collection, as well as image projection technology, can be used in the LED text and graphic display for space-constrained product design [12]. In recent years, the capacitive touchpad is very popular, low-cost human-computer interface. Its mechanical structure is not easy to wear, convenient for finger operation, has become the mainstream operating interface of information products. The

sensing method is to measure the amount of change of the capacitance of the panel to determine the detection of finger movements and provide the operation control interface when the fan is running. The common paper-based touchpad structure, the circuit is printed directly on the paper substrate, forming a thin layer of resistance, simple structure and easy to manufacture. It can be applied to surface feature products with flexibility and durability, and can be used as the ultra-thin, lightweight and lower cost portable product design [13]. If further through the IR sensor technology, can develop contactless sensing feature for fan remote control [14]. If the future in the development of portable or wearable electronic product applications, because it is human-centered, with special emphasis on the product can be self-powered, as well as operating flexibility and other characteristics. And, no matter how the material of the touch object, by the capacitive touch sensing technology, from the finger touch to produce impedance changes. Therefore, in the sensing circuit design for thin touchpad, to be able to generate a stable voltage signal, set control and program logic judgments [15].

In this study, the interactive display structure on the impeller is equipped with a power filter circuit which includes sensors and LED chips for the display control purpose. For users, this structure provides an intuitive way of displaying interactive information. This type of power filter circuit has a bridge rectifier inside so that the AC power generated by the alternator inside the impeller rotor can be converted into DC power. In addition, potential surges are filtered out so that this structure can serve as a stable voltage source.

For the design of the magnetic induction sensor, its main function is to provide an operating reference point for the configuration and analysis of the operating condition for the display program. The proposed design can stabilize the display status of texts or graphics so that the display is not flickering during operation. For the control of the LED chips, the design considerations include the layout of LED chips and the limitation on the definition of the texts or graphics. Control signals are transmitted from the output pins of the MCU to those 8 LED chips while the 9th LED is connected in parallel to the 8th one. The built-in firmware is programmed to control the on/off sequence of the LED chips. The information such as texts or graphics can be displayed by the phenomenon of persistence of vision with variations in the operating model.

Moreover, a touchpad of the capacitive touch sensing technology is also included in the main program of this design for controlling the operation of the circuits and input/output models. The input model deals with receiving the signals from the magnet sensors in order to determine the operation sequence of the program and the output signals to be sent to the LED chips as shown in Fig. 4.

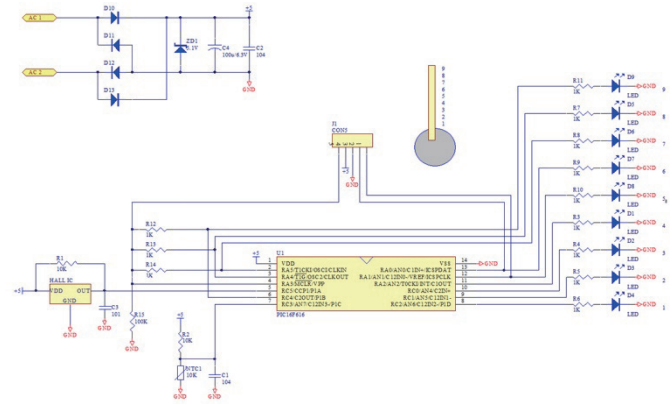


Fig. 4 Interactive display circuit design.

Experimental Results

The results of the experiment in this study indicated that, the default texts and graphics are correctly displayed on the fan during operation. In comparison, there is a potential risk that a user might hurt his/her figure by a conventional cooling fan if it does not come with a fan grille. The design in this study includes a capacitive touchpad sensor on the impeller. This sensor can detect the variation when a user touches the fan grille gently and changes the display information such as temperature or fan speed as shown in Fig. 5. Since the motor is relocated to the outer region of the fan structure, diversified fan designs can be realized so that a conventional cooling fan can be transformed into a fashion product by including cultural creativity. From the standpoint of product modularization and diversified designs, a product family can also be developed. Based on this condition, a derived design of the 4" Mini LED Desk Fan is also created in this study by implementing the built-in power generating module. By thinking outside the box, product functions can be redefined and a cooling fan is no long just a functional cooling component. These types of products as shown in Fig. 6 can better fit the niche market and therefore the value of a product is increased substantially.

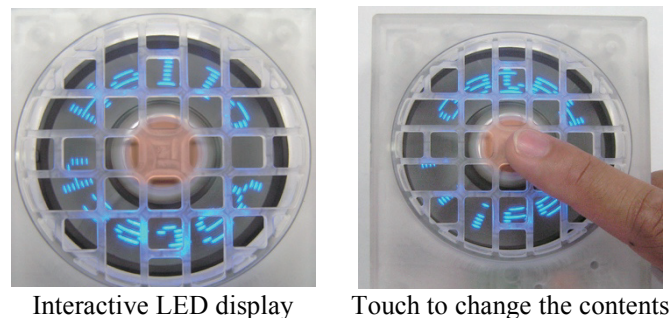


Fig. 5 Interactive LED display test.

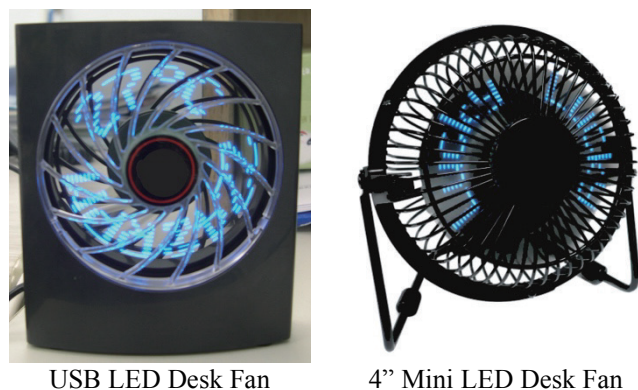


Fig. 6 LED display module applications.

Conclusions

In this study, the technologies of external driving motor, dynamic LED text and graphics display are integrated into the built-in power generating module of a cooling fan in order to realize a contactless design with self-generate power. No external power is required. The structure is simple and the thickness of a fan can also be reduced. The feasibility of controlling information such as texts, graphics, temperature, and fan speed by the firmware within the MCU is also investigated. The proposed structure can realize the integration with the creativity of digital contents so that the added value of a cooling fan is greatly enhanced. From the standpoint of energy conservation, the internal power generating module of this fan can transform the motor into a power generator for supplying the power required by the LED chips. This design meets the energy-efficient requirement. In addition, the external driving motor technology can greatly reduce the resistance that is caused by a motor at the fan hub. The thermal performance can be enhanced effectively due to the optimization of the flow field. In addition, the noise level is also reduced due to the fact that this design can operate at a lower speed. Therefore, this design can be used in an environment that is noise sensitive and needs to be quiet. For follow-up studies, this design can be implemented to the product designs including thermometers, clocks, and exhaust fans. This study realized the design and development of diversified products by integrating a new motor structure with new LED and sensor technologies. The result of this study can serve as a good reference for future industrial upgrading and transforming plans.

References

- [1] H. Z. Lu, L. Huang, R. M. C. So and J. Wang, "A computational study of the interaction noise from a small axial-flow fan," *Journal of the Acoustical Society of America*, 2007, 122(3), pp. 1404-1415.
- [2] H. J. Lee, S. U. Chung, and S. M. Hwang, "Acoustic

- resonance of outer-rotor brushless dc motor for air-conditioner fan," *Journal of Applied Physics*, 2008, 103(7), 07F116.
- [3] N. F. Zulkarnain, T. Ibrahim, and M. F. Romlie, "Design and optimization of permanent magnet machine for ceiling fan," In *Intelligent and Advanced Systems (ICIAS)*, 2016, 6th International Conference on (pp. 1-6), IEEE.
- [4] Y. Kai and F. Yaojing, "Design of novel spiral magnetic poles and axial-cooling structure of outer-rotor PM torque motor," *IEEE Transactions on Applied Superconductivity*, 2010, 20(3), pp. 838-841.
- [5] A. M. El-Refaie, J. P. Alexander, S. Galioto, P. B. Reddy, K. Huh, P. de Bock and X. Shen, "Advanced high-power-density interior permanent magnet motor for traction applications," *IEEE Transactions on Industry Applications*, 2014, 50(5), pp. 3235-3248.
- [6] A. Lelkes, J. Krotsch and R. W. De Doncker, "Low-noise external rotor BLDC motor for fan applications," In *Industry Applications Conference*, 2002, 37th IAS Annual Meeting. Conference Record of the (Vol. 3, pp. 2036-2042), IEEE.
- [7] M. C. Tsai, M. H. Weng and M. F. Hsieh, "Computer-aided design and analysis of new fan motors," *IEEE transactions on magnetics*, 2002, 38(5), pp. 3467-3474.
- [8] J. F. Gieras, D. Chojnowski and P. Mikulski, "Analysis of steady-state and transient performance of two-phase PM motors for computer fans," In *Electric Machines & Drives Conference (IEMDC)*, 2015, IEEE International (pp. 624-629), IEEE.
- [9] H. C. Su and H. Y. Xu, "Cooling ability of heat sinks with oscillating-fan models using electromagnetic force," In *Thermal Measurement, Modeling & Management Symposium (SEMI-THERM)*, 2016, 32nd (pp. 7-14), IEEE.
- [10] J. Y. Lee, L. Y. Huang and C. Y. Chen, "Design and implementation of contactless maglev rotating power transfer system with new rotary inductive coupled structure," In *Power Electronics and Motion Control Conference (IPEMC-ECCE Asia)*, 2016, IEEE 8th International (pp. 2442-2449), IEEE.
- [11] A. Flammini, D. Marioli, E. Sardini and M. Serpelloni, "An autonomous sensor with energy harvesting capability for airflow speed measurements. In *Instrumentation and Measurement Technology Conference (I2MTC)*, 2010, IEEE (pp. 892-897), IEEE.
- [12] X. Fan, W. D. Leon-Salas, T. Fischer and A. Perez-Olvera, "An LED-based image sensor with energy harvesting and projection capabilities," In *SENSORS*, 2016, IEEE (pp. 1-3), IEEE.
- [13] R. Z. Li, A. Hu, T. Zhang and K. D. Oakes, "Direct writing on paper of foldable capacitive touch pads with silver nanowire inks," *ACS applied materials & interfaces*, 2014, 6(23), pp. 21721-21729.
- [14] C. T. Chuang, T. Chang, P. H. Jau and F. R. Chang, "Touchless positioning system using infrared LED sensors," In *System Science and Engineering (ICSSE)*, 2014, IEEE International Conference on (pp. 261-266), IEEE.
- [15] X. Z. Jiang, Y. J. Sun, Z. Fan and T. Y. Zhang, "Integrated flexible, waterproof, transparent, and self-powered tactile sensing panel," *ACS nano*, 2016, 10(8), pp. 7696-7704.